CHIRAL CHERN–WEIL THEORY

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Abstract. Classical Chern–Weil theory extracts characteristic classes from the curvature of a principal bundle using invariant polynomials. We develop a chiral analogue in the setting of modular Koszul duality for vertex algebras. The ordered bar complex $B^{\mathrm{ord}}(\mathcal{A})$ over the Ran space of a curve X plays the role of a chiral classifying space; the Maurer–Cartan element $\Theta^{E_1}_{\mathcal{A}}$ plays the role of a universal connection, with degree-2 component the classical r-matrix $r(z)$. The averaging comparison from the ordered to the symmetric convolution algebra plays the role of the invariant polynomial where the descent datum exists: it kills the E_1 operadic indices, and the scalar shadow data is then extracted by family-specific moment-residue, Sugawara, and composite-sector rules. As the curve X varies over $\overline{\mathcal{M}}_g$, the fiberwise bar differential has a fiber Arakelov curvature term, while the base obstruction class is $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A})\lambda_g$. The degree-by-degree images $(\kappa, \mathfrak{C}, \mathfrak{Q}, \ldots) = \mathrm{av}(r(z), r_3, r_4, \ldots)$ form the shadow tower. The four standard depth types (G/L/C/M) are read from the tuple $(S_3, S_4, \text{termination})$, not from the single downstream discriminant $\Delta = 8\kappa S_4$. For multi-weight algebras, the scalar formula conditionally acquires a cross-channel correction $\delta F_g^{\mathrm{cross}}$, which we compute explicitly for $\mathcal{W}_3$ at genus 2: $\delta F_2(\mathcal{W}_3) = (c+204)/(16c)$.

Contents

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1. Introduction

1.1. **The deficiency.** The bar complex of a chiral algebra $\mathcal{A}$ on a curve X varies as X moves in the moduli space $\overline{\mathcal{M}}_g$. At genus 0, the bar differential squares to zero. At genus 1, it does not: the fiberwise differential acquires fiber Arakelov curvature, and its pushforward to the base produces the Hodge-class obstruction with coefficient κ.

This curvature carries E_1 operadic indices: for an affine Kac–Moody algebra $V_k(\mathfrak{g})$, the r-matrix $r(z) = k\,\Omega/z$ is valued in $\mathfrak{g} \otimes \mathfrak{g}$, and the scalar κ is the trace of a tensor. For the Virasoro algebra at central charge c, the r-matrix $r(z) = (c/2)/z^3 + 2T/z$ mixes c-number and field-dependent terms, and the scalar $\kappa = c/2$ is the vacuum residue after a weight-dependent moment shift.

The question is: what mechanism extracts the scalar shadow data from $r(z)$? The answer has a precise classical analogue. In classical Chern–Weil theory, the curvature of a principal G-bundle is $\mathfrak{g}$-valued, and a G-invariant polynomial kills the internal indices to produce a scalar characteristic class. The chiral analogue replaces the invariant polynomial by the *averaging map* $\mathrm{av}\colon \mathfrak{g}^{E_1}_{\mathcal{A}} \to \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ (the Σ_n-coinvariant comparison on the descent locus), which kills the E_1 operadic indices. The final scalar extraction is family-specific.

1.2. **Results.** The main contributions are:

(1) A systematic parallel between classical Chern–Weil theory and the modular Koszul duality programme (Sections 2–5).
(2) Explicit computation of the averaging map on the r-matrices of all standard families: Heisenberg, Virasoro, affine Kac–Moody (all types A–G), principal $\mathcal{W}_N$, and free-field systems (Section 4).
(3) Identification of the shadow tower $(\kappa, \mathfrak{C}, \mathfrak{Q}, \ldots)$ as the sequence of chiral characteristic classes, with the G/L/C/M classification as the classification of chiral classifying spaces by the type of their characteristic ring (Section 6).
(4) Derivation and explicit computation of the multi-weight cross-channel correction $\delta F_g^{\mathrm{cross}}$, including the first closed-form result $\delta F_2(\mathcal{W}_3) = (c+204)/(16c)$ (Section 7).

2. Classical Chern–Weil theory

2.1. **Connections and curvature.** Let G be a Lie group with Lie algebra $\mathfrak{g}$, and let $P \to M$ be a principal G-bundle over a smooth manifold M. The bundle may be topologically non-trivial, but any single local trivialization sees only $\mathfrak{g}$. To detect the global topology from local data, one needs a device that is defined locally but transforms coherently: a connection.

Definition 2.1. A *connection* on P is a $\mathfrak{g}$-valued 1-form $\nabla \in \Omega^1(P; \mathfrak{g})$ satisfying:

(i) $R_g^* \nabla = \mathrm{Ad}(g^{-1})\nabla$ for all $g \in G$ (equivariance);
(ii) $\nabla(\xi_P) = \xi$ for all $\xi \in \mathfrak{g}$ (normalization along fibers),

where ξ_P is the fundamental vector field of ξ and R_g is the right G-action on P. In a local trivialization over $U \subset M$, a connection is represented by $A \in \Omega^1(U; \mathfrak{g})$.

Definition 2.2. The *curvature* of a connection ∇ is the $\mathfrak{g}$-valued 2-form

$$F_\nabla = d\nabla + \tfrac{1}{2}[\nabla, \nabla] \;\in\; \Omega^2(M; \mathrm{Ad}(P)).$$

In local coordinates: $F = dA + A \wedge A$.

The curvature measures the failure of parallel transport around infinitesimal loops to be trivial. It is $\mathfrak{g}$-valued: it carries internal indices from the Lie algebra.

Lemma 2.3 (Bianchi identity). $d_\nabla F_\nabla = 0$.

Proof. Compute $d_\nabla F = d_\nabla(d\nabla + \frac{1}{2}[\nabla, \nabla]) = d^2\nabla + \frac{1}{2}[d_\nabla \nabla, \nabla] - \frac{1}{2}[\nabla, d_\nabla \nabla]$. Using $d^2 = 0$, $d_\nabla \nabla = F$, and the Jacobi identity for $\mathfrak{g}$, all terms cancel. □

2.2. **The Chern–Weil homomorphism.**

Definition 2.4. Let $\mathrm{Sym}^k(\mathfrak{g}^*)^G$ denote the space of G-invariant symmetric k-linear forms on $\mathfrak{g}$. For $P \in \mathrm{Sym}^k(\mathfrak{g}^*)^G$, the *Chern–Weil form* is

$$\mathrm{CW}(P) = P(\underbrace{F_\nabla, \ldots, F_\nabla}_{k}) \;\in\; \Omega^{2k}(M).$$

The form $\mathrm{CW}(P)$ is scalar-valued: the invariant polynomial contracts all $\mathfrak{g}$-indices. Its construction requires the curvature to carry indices (otherwise there is nothing to contract) and the polynomial to be invariant (otherwise the result would depend on the trivialization).

Theorem 2.5 (Chern–Weil). (i) $\mathrm{CW}(P)$ *is closed:* $d[\mathrm{CW}(P)] = 0$.
(ii) *The cohomology class* $[\mathrm{CW}(P)] \in H^{2k}(M; \mathbb{R})$ *is independent of the connection* ∇.
(iii) *The resulting map* $\mathrm{CW}\colon \mathrm{Sym}^*(\mathfrak{g}^*)^G \to H^*(M; \mathbb{R})$ *is a ring homomorphism.*

Proof. (i) The Bianchi identity $d_\nabla F = 0$ and the G-invariance of P imply $d[P(F^k)] = k\,P(d_\nabla F, F^{k-1}) = 0$.
(ii) Let ∇_0, ∇_1 be connections with curvatures F_0, F_1. Set $\alpha = \nabla_1 - \nabla_0 \in \Omega^1(M; \mathrm{Ad}(P))$ and interpolate: $\nabla_t = \nabla_0 + t\alpha$, $F_t = F_0 + t\, d_{\nabla_0}\alpha + t^2 \alpha \wedge \alpha$. Then

$$P(F_1^k) - P(F_0^k) = d \int_0^1 k\, P(\alpha, F_t^{k-1})\, dt.$$

The Chern–Simons transgression $\mathrm{TP} = \int_0^1 k\,P(\alpha, F_t^{k-1})\,dt$ exhibits the difference as exact.
(iii) Multiplicativity of the wedge product. □

Example 2.6 (Abelian: $G = U(1)$)**.** $\mathfrak{g} = \mathbb{R}$, $F = dA$ is already scalar. The unique invariant polynomial (up to scale) is the identity. The first Chern class $c_1 = \frac{1}{2\pi i}[F] \in H^2(M; \mathbb{Z})$ is the sole characteristic class. The classifying space $BU(1) = \mathbb{CP}^\infty$ has $H^*(BU(1)) = \mathbb{Z}[c_1]$.

Example 2.7 ($G = SU(2)$)**.** $\mathfrak{g} = \mathfrak{su}(2)$ is traceless, so $c_1 = 0$. The first non-trivial invariant polynomial is $P(X, Y) = -\frac{1}{8\pi^2}\operatorname{tr}(XY)$, giving the second Chern class $c_2 = -\frac{1}{8\pi^2}[\operatorname{tr}(F \wedge F)]$. On S^4, $\int_{S^4} c_2 \in \mathbb{Z}$ is the instanton number.

Remark 2.8 (Structure of the invariant ring). For $G = GL_n(\mathbb{C})$, the invariant polynomials are generated by $P_k(X) = \operatorname{tr}(X^k)$, $k = 1, \ldots, n$, producing Chern classes $c_1, \ldots, c_n$. The number of independent generators of $\mathrm{Sym}^*(\mathfrak{g}^*)^G$ determines the *type* of the classifying space BG.

3. Chiral algebras, the bar complex, and the r-matrix

Classical Chern–Weil theory requires three ingredients: a connection, a curvature, and an invariant polynomial. Each has a chiral counterpart, and each requires construction. The connection is the Maurer–Cartan element of the E_1 convolution algebra; the curvature at degree 2 is the r-matrix; the invariant polynomial is the averaging map. This section constructs the first two.

3.1. **Chiral algebras and OPE.** Let X be a smooth algebraic curve. A chiral algebra $\mathcal{A}$ on X is a factorization algebra encoding operator product expansions. In local coordinates, the data consists of strong generators $\varphi_1, \ldots, \varphi_r$ of conformal weights $h_1, \ldots, h_r$, with operator product expansion

$$\varphi_i(z)\,\varphi_j(w) \;\sim\; \sum_{p=1}^{h_i+h_j} \frac{c_{ij}^p(w)}{(z-w)^p}$$

where $c_{ij}^p(w)$ are OPE coefficients, possibly depending on the remaining fields.

3.2. **The ordered bar complex.**

Definition 3.1. The *ordered bar complex* of $\mathcal{A}$ is

$$B^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$$

where $\bar{\mathcal{A}} = \ker(\varepsilon)$ is the augmentation ideal of $\mathcal{A}$, s^{-1} is the desuspension ($|s^{-1}v| = |v| - 1$), and T^c is the cofree conilpotent coalgebra with the deconcatenation coproduct. Grading is cohomological ($|d| = +1$).

Warning 3.2. Three distinct errors are common in the bar-complex formula. First, using $\mathcal{A}$ instead of $\bar{\mathcal{A}}$ (the augmentation ideal is essential). Second, using the suspension s instead of desuspension s^{-1} (bar is down). Third, using the tensor algebra T instead of the cofree coalgebra T^c (the deconcatenation coproduct is the coalgebra structure).

On a curve X, the bar complex is a factorization coalgebra on the Ran space $\mathrm{Ran}(X) = \bigsqcup_{n\geq 1} \mathrm{Conf}_n(X)$. The bar differential uses the logarithmic propagator $\eta_{12} = d\log(z_1 - z_2)$ at genus 0.

3.3. **The r-matrix from the OPE.** The bar differential at degree 2 encodes the collision of two generators. The propagator $d\log(z_1 - z_2)$ absorbs one pole from the OPE, giving the *r-matrix*:

Definition 3.3. The *r-matrix* of $\mathcal{A}$ is the degree-2 component of the E_1 Maurer–Cartan element $\Theta^{E_1}_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{E_1}_{\mathcal{A}})$:

$$r(z) \;=\; (\Theta^{E_1}_{\mathcal{A}})_2 \;\in\; (\mathfrak{g}^{E_1}_{\mathcal{A}})_2.$$

It is obtained from the OPE data by $d\log$-absorption: the pole order of r equals $p_{\max}(\mathrm{OPE}) - 1$.

We now compute the r-matrix for every standard family.

Computation 3.4 (Heisenberg at level k)**.** Generator J, weight $h = 1$. OPE:

$$J(z)J(w) = \frac{k}{(z-w)^2}.$$

Pole order 2. After $d\log$-absorption:

$$\boxed{r^{\mathrm{Heis}}(z) = \frac{k}{z}.}$$

Simple pole. At $k = 0$: $r = 0$ (trivial level).

Computation 3.5 (Virasoro at central charge c)**.** Generator T, weight $h = 2$. OPE:

$$T(z)T(w) = \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}.$$

Pole order 4. After $d\log$-absorption, only the singular part of the OPE contributes to the collision residue; the regular ∂T term drops (its $d\log$ coefficient vanishes identically). Thus:

$$\boxed{r^{\mathrm{Vir}}(z) \;=\; \frac{c/2}{z^3} \;+\; \frac{2T}{z}.}$$

Cubic pole, *not* quartic. The leading term $(c/2)/z^3$ is a c-number; the subleading term $2T/z$ is field-dependent. The regular term ∂T from the OPE is absent in r^{Vir}: it is $d\log$-absorbed, or equivalently it is a total derivative on the configuration space and contributes 0 to the residue pairing. At central charge $c = 0$, the vacuum cubic term vanishes and the field-dependent piece $2T/z$ remains.

Computation 3.6 (Affine Kac–Moody $V_k(\mathfrak{g})$)**.** Generators J^a, $a = 1, \ldots, \dim(\mathfrak{g})$, all weight $h = 1$. Let $(\ ,\ )$ be the normalized invariant form (long roots squared length 2), $\{t^a\}$ an orthonormal basis. OPE:

$$J^a(z)J^b(w) = \frac{k\,\delta^{ab}}{(z-w)^2} + \frac{f^{ab}{}_c\,J^c(w)}{z-w}.$$

After $d\log$-absorption:

$$\boxed{r^{\mathrm{KM}}(z) = \frac{k\,\Omega}{z}, \qquad \Omega = \sum_a t^a \otimes t^a \in \mathfrak{g} \otimes \mathfrak{g}.}$$

This is the *trace-form convention*. The level k survives $d\log$-absorption. At $k = 0$: $r = 0$. In the *Kazhdan–Lusztig (KZ) convention*: $r^{\mathrm{KZ}}(z) = \Omega_{\mathrm{KZ}}/((k + h^\vee)z)$ where Ω_{KZ} is the Killing-normalized Casimir; at $k = 0$ this gives $\Omega_{\mathrm{KZ}}/(h^\vee z) \neq 0$ for non-abelian $\mathfrak{g}$ (the Lie bracket persists). The two conventions are *not equal* as rational functions of k: they are related by a level-dependent rescaling $\Omega_{\mathrm{tr}} = (2h^\vee)^{-1}\Omega_{\mathrm{KZ}}$ together with $k \mapsto k(k + h^\vee)/(2h^\vee)$, or equivalently by absorbing the $(k + h^\vee)^{-1}$ factor into the KZ flat connection. The equation $k\,\Omega_{\mathrm{tr}} = \Omega_{\mathrm{KZ}}/(k + h^\vee)$ should be read as a structural *change of basis* between conventions, not as an identity of rational functions in k: at $k = 0$ the LHS vanishes while the

RHS is finite and non-zero. We adopt the trace convention throughout this note; every r-matrix in this section satisfies $r|_{k=0} = 0$, exhibiting cleanly.

Computation 3.7 (Principal $\mathcal{W}_N$ algebra)**.** Generators $W^{(s)}$ at weights $s = 2, 3, \dots, N$ ($N - 1$ generators). The self-OPE of the spin-s primary in the Fateev–Lukyanov normalization has leading singularity

$$W^{(s)}(z)\, W^{(s)}(w) = \frac{c/s}{(z-w)^{2s}} + \cdots$$

After d log-absorption, the vacuum leading term is

$$r^{\mathrm{vac}}_{ss}(z) = \frac{c/s}{z^{2s-1}} + \cdots$$

The cross-OPE $W^{(s_1)}(z)\, W^{(s_2)}(w)$ for $s_1 \neq s_2$ has leading coefficient that is field-dependent, not a c-number.

Computation 3.8 (Fermionic bc system at weight λ)**.** Generators: b (weight λ) and c (weight $1-\lambda$), both fermionic. Central charge $c_{bc}(\lambda) = 1 - 3(2\lambda - 1)^2$. The singular OPEs:

$$b(z)\, c(w) \sim \frac{1}{z-w}, \qquad b(z)\, b(w) \sim 0, \qquad c(z)\, c(w) \sim 0.$$

The only singular OPE has a simple pole (pole order 1). After d log-absorption the cross-contraction has pole order 0. It does not contribute to the scalar collision residue:

$$r^{\mathrm{coll}}_{bc}(z) = 0.$$

The self-OPEs $b(z)b(w)$ and $c(z)c(w)$ are regular (fermionic statistics forbid self-contraction at generic λ), so $r_{bb} = r_{cc} = 0$.

At $\lambda = 2$: the reparametrization ghost system of the bosonic string, with $c_{bc}(2) = -26$. At $\lambda = 1/2$: a single Dirac fermion, $c_{bc}(1/2) = 1$.

Computation 3.9 (Bosonic $\beta\gamma$ system at weight λ)**.** Generators: β (weight λ) and γ (weight $1-\lambda$), both bosonic. Central charge $c_{\beta\gamma}(\lambda) = 2(6\lambda^2 - 6\lambda + 1) = -c_{bc}(\lambda)$. OPE:

$$\beta(z)\, \gamma(w) \sim \frac{1}{z-w}.$$

After d log-absorption the cross-contraction has pole order 0, so $r^{\mathrm{coll}}_{\beta\gamma}(z) = 0$ in the collision-residue shadow. The modular characteristic comes from the stress tensor and composite sector.

Complementarity check: $c_{\beta\gamma}(\lambda) + c_{bc}(\lambda) = 0$ for all λ. At $\lambda = 1$: $c_{\beta\gamma}(1) = 2$, $c_{bc}(1) = -2$, sum $= 0$. At $\lambda = 2$: $c_{\beta\gamma}(2) = 26$, $c_{bc}(2) = -26$, sum $= 0$ (bosonic string ghost cancellation).

4. The averaging map and the modular characteristic

4.1. Definition of the averaging map.

Definition 4.1 (Averaging comparison)**.** An *averaging datum* is an R-twisted descent datum from the ordered (E_1) convolution Lie algebra to the modular symmetric convolution Lie algebra. With such a datum fixed, the arity-n comparison is the quotient to Σ_n-coinvariants:

$$\mathrm{av}\colon \mathfrak{g}^{E_1}_{\mathcal{A}} \longrightarrow \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}.$$

For finite-dimensional aritywise calculations over characteristic zero, the Reynolds operator

$$\mathrm{Reyn}(\phi)(g,n) = \frac{1}{n!} \sum_{\sigma \in \Sigma_n} \sigma \cdot \phi(g,n)$$

identifies invariants with coinvariants after a choice of splitting; it is not a canonical dg Lie section.

Theorem 4.2 (Averaging on the descent locus). *On the descent locus, the comparison map satisfies:*

(i) av *is a dg Lie morphism.*
(ii) $\mathrm{av}(\Theta^{E_1}_{\mathcal{A}}) = \Theta_{\mathcal{A}}$*: the ordered MC element maps to the symmetric MC element.*
(iii) $\ker(\mathrm{av})$ *records the R-matrix structure, the braiding, and the Yangian coherences invisible to the symmetric shadow.*

At the level of MC elements: the ordered MC element has degree-wise components $(r(z), r_3, r_4, \ldots)$, where $r(z)$ is the classical r-matrix (degree 2), r_3 is the KZ associator (degree 3), and r_4 is the quartic R-matrix coherence (degree 4). These satisfy the Stasheff associahedra boundary equations: the classical Yang–Baxter equation for $r(z)$, the pentagon for r_3, the quartic identity for r_4.

The modular MC element $\Theta_{\mathcal{A}} = \mathrm{av}(\Theta^{E_1}_{\mathcal{A}})$ retains only the Σ_n-invariant part of each degree.

4.2. **The Σ_2 action on an r-matrix.** Before computing κ for specific families, we work out what the averaging map does to a generic r-matrix at degree 2, step by step.

At degree 2, the E_1 convolution algebra consists of elements on the ordered configuration space $\mathrm{Conf}_2(X) = \{(z_1, z_2) \in X^2 : z_1 \neq z_2\}$. Let $z = z_1 - z_2$ be the relative coordinate. The r-matrix is a section of $\mathrm{End}(V)^{\otimes 2}$ over $\mathrm{Conf}_2(X)$, where V is the space of strong generators. For generators φ_i, φ_j, write

$$r(z) = \sum_{i,j} r_{ij}(z) \cdot e_i \otimes e_j \;\in\; \mathrm{End}(V)^{\otimes 2} \otimes \mathbb{C}((z)),$$

where $\{e_i\}$ is a basis for V and each $r_{ij}(z)$ is a Laurent series in z (possibly with field-dependent coefficients).

The group Σ_2 acts on $\mathrm{Conf}_2(X)$ by swapping: $(z_1, z_2) \mapsto (z_2, z_1)$, equivalently $z \mapsto -z$. Simultaneously, Σ_2 acts on $\mathrm{End}(V)^{\otimes 2}$ by swapping tensor factors: $e_i \otimes e_j \mapsto e_j \otimes e_i$. The combined action on the r-matrix is:

$$(\sigma \cdot r)(z) = \sum_{i,j} r_{ij}(-z) \cdot e_j \otimes e_i.$$

The Σ_2-coinvariant projection is:

$$\mathrm{av}_2(r)(z) = \tfrac{1}{2}\big(r(z) + \sigma \cdot r(z)\big) = \tfrac{1}{2} \sum_{i,j} \big(r_{ij}(z) + r_{ji}(-z)\big) \cdot e_i \otimes e_j. \tag{4.1}$$

4.3. **From function to scalar: the cyclic trace.** The averaged element (4.1) is still a function of z with values in $\mathrm{End}(V)^{\otimes 2}$. Two further operations extract a scalar.

Residue extraction. The modular convolution algebra at degree 2 pairs with the propagator $\eta_{12} = d\log(z_1 - z_2)$, which extracts the residue at $z = 0$. The relevant coefficient is

$$\mathrm{Res}_{z=0}\big[\mathrm{av}_2(r)(z)\big] = \tfrac{1}{2} \sum_{i,j} \big(\mathrm{Res}_{z=0}\, r_{ij}(z) + \mathrm{Res}_{z=0}\, r_{ji}(-z)\big) \cdot e_i \otimes e_j.$$

For an even-order pole $r_{ij}(z) = c/z^{2m}$: $\mathrm{Res}_{z=0}(c/z^{2m}) = 0$ and $\mathrm{Res}_{z=0}(c/(-z)^{2m}) = 0$. For an odd-order pole $r_{ij}(z) = c/z^{2m+1}$: $\mathrm{Res}_{z=0}(c/z^{2m+1}) = 0$ unless $m = 0$ (simple pole), and $\mathrm{Res}_{z=0}(c/(-z)^{2m+1}) = (-1)^{2m+1}\mathrm{Res}_{z=0}(c/z^{2m+1}) = -\mathrm{Res}_{z=0}(c/z^{2m+1})$.

For a simple-pole r-matrix $r_{ij}(z) = c_{ij}/z$:

$$\mathrm{Res}_{z=0}\, c_{ij}/z + \mathrm{Res}_{z=0}\, c_{ji}/(-z) = c_{ij} - c_{ji}.$$

This is the *antisymmetric* part of the coefficient matrix. For a symmetric coefficient ($c_{ij} = c_{ji}$, as for $k\,\delta^{ab}$ in the Heisenberg/KM double pole), the two terms cancel!

The resolution: for higher-weight generators, the relevant coefficient is not the simple residue but the *moment residue*, as we now explain.

The moment and the cyclic trace. The cyclic trace on $\mathrm{End}(V^{\otimes 2})$ incorporates the conformal weights of the generators. For a generator φ_i of weight h_i, the OPE pairing carries a weight factor z^{2h_i-2} from the conformal Ward identity: the two-point function $\langle\varphi_i(z)\varphi_i(0)\rangle \propto z^{-2h_i}$ contributes a volume factor z^{2h_i}, and the propagator $d\log z$ contributes z^{-2}, giving a net factor z^{2h_i-2}.

The cyclic trace therefore evaluates the $(2h_i-2)$*-th moment* of the residue:

$$\mathrm{tr}_{\mathrm{cyc}}\big[\mathrm{Res}_{z=0}\, r(z)\big] = \sum_i \mathrm{Res}_{z=0}\big[z^{2h_i-2}\cdot r^{\mathrm{vac}}_{ii}(z)\big] + (\text{Sugawara shift}).$$

For a single generator of weight h with vacuum r-matrix $r^{\mathrm{vac}}(z) = c_h/z^{2h-1}+\cdots$ (leading vacuum pole), the moment residue is:

$$\mathrm{Res}_{z=0}\big[z^{2h-2}\cdot c_h/z^{2h-1}\big] = \mathrm{Res}_{z=0}\big[c_h/z\big] = c_h.$$

The moment factor z^{2h-2} exactly cancels the excess pole order, reducing any leading pole to a simple pole whose residue is the vacuum OPE coefficient c_h.

Field-dependent terms. The full r-matrix contains field-dependent terms in addition to the vacuum (c-number) part. For Virasoro: $r(z) = (c/2)/z^3 + 2T/z$, where $2T/z$ is field-dependent.

The vacuum projection $r^{\mathrm{vac}}(z) = \langle 0|r(z)|0\rangle$ retains only the c-number part. The cyclic trace annihilates field-dependent terms because they have zero vacuum expectation value: $\langle 0|T|0\rangle = 0$. More precisely, the trace on $\mathrm{End}(V^{\otimes 2})$ contracts the two V-slots against the vacuum state, and any term proportional to a non-identity field gives zero after contraction.

For Virasoro at $h=2$:

$$\mathrm{Res}_{z=0}\big[z^2\cdot r(z)\big] = \mathrm{Res}_{z=0}\big[z^2\cdot\big((c/2)/z^3+2T/z\big)\big] = \mathrm{Res}_{z=0}\big[(c/2)/z + 2Tz\big] = c/2 + 0.$$

The field-dependent term $2T$ is regular at $z=0$ after the moment shift, so their residue vanishes. This is a general mechanism: the moment factor z^{2h-2} shifts the vacuum pole to a simple pole and simultaneously makes all field-dependent terms regular.

For families whose scalar shadow is purely diagonal, the extraction rule is

$$\kappa(\mathcal{A}) = \sum_i \kappa_i + \kappa_{\mathrm{sp}}, \tag{4.2}$$

where $\kappa_i = \mathrm{Res}_{z=0}[z^{2h_i-2}\cdot r^{\mathrm{vac}}_{ii}(z)]$ is the per-channel diagonal contribution and κ_{sp} is the Sugawara shift from simple-pole self-contraction (nonzero only for non-abelian same-weight generators). First-order systems and non-principal reductions also contain composite-sector contributions, so (4.2) is an extraction rule on the diagonal standard families, not a universal trace-residue theorem.

4.4. **Explicit trace: $\mathfrak{sl}_2$ at level k.** We carry out the full computation for $\mathfrak{g} = \mathfrak{sl}_2$, $\dim = 3$, $h^\vee = 2$. The standard basis is

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0\end{pmatrix},\quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0\end{pmatrix},\quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1\end{pmatrix},$$

with bracket $[e,f]=h$, $[h,e]=2e$, $[h,f]=-2f$.

The normalized invariant form (long roots squared length 2) is $(x,y) = \frac{1}{2h^\vee}\mathrm{tr}_{\mathrm{ad}}(\mathrm{ad}(x)\,\mathrm{ad}(y)) = \frac{1}{4}\mathrm{tr}_{\mathrm{ad}}(\mathrm{ad}(x)\,\mathrm{ad}(y))$. Compute: $(e,f)=1$, $(h,h)=2$, $(e,e)=(f,f)=0$. The form is indefinite; no real orthonormal basis exists.

The Casimir is $\Omega = \sum_a t^a\otimes t_a$ where $\{t^a\}$ and $\{t_a\}$ are dual bases for $(\ ,\ )$. With $t^1=e$, $t^2=f$, $t^3=h/2$ and $t_1=f$, $t_2=e$, $t_3=h/2$ (so that $(t^a,t_b)=\delta^a_b$):

$$\Omega = e\otimes f + f\otimes e + \tfrac{1}{2}\,h\otimes h.$$

Check: $\mathrm{tr}(\Omega) = (e,f)+(f,e)+\frac{1}{2}(h,h) = 1+1+1 = 3 = \dim(\mathfrak{sl}_2)$.

The r-matrix is $r(z) = k\,\Omega/z$.

Double-pole channel. Each generator J^a has weight $h = 1$; the moment factor is $z^0 = 1$. The double-pole OPE $J^a(z)\,J_a(w) \sim k\,(t^a, t_a)/(z-w)^2 = k/(z-w)^2$ gives r-matrix k/z per dual pair. Three dual pairs, total double-pole trace:

$$\mathrm{tr}\big(d^2_{\mathrm{fib}}\big|_{\mathrm{dp}}\big) = 3k \cdot \omega_1.$$

Normalizing by $2h^\vee = 4$: $\kappa_{\mathrm{dp}} = 3k/(2h^\vee) = 3k/4$.

Simple-pole channel. Use the normalized invariant form with dual bases $\{x_i\}$ and $\{x^i\}$, so

$$\Omega = \sum_i x_i \otimes x^i = e \otimes f + f \otimes e + \tfrac{1}{2} h \otimes h.$$

The adjoint Casimir identity is

$$\sum_i [x_i, [x^i, y]] = 2h^\vee y.$$

For $\mathfrak{sl}_2$, $h^\vee = 2$; with $x_i = e, f, h$ and $x^i = f, e, h/2$, the identity gives $\sum_i [x_i, [x^i, e]] = 4e$. This is the simple-pole self-contraction responsible for the Sugawara shift.

The Sugawara trace:

$$\kappa_{\mathrm{sp}} = \frac{\dim(\mathfrak{g})}{2} = \frac{3}{2}.$$

Total.

$$\kappa(V_k(\mathfrak{sl}_2)) = \kappa_{\mathrm{dp}} + \kappa_{\mathrm{sp}} = \frac{3k}{4} + \frac{3}{2} = \frac{3(k+2)}{4} = \frac{\dim(\mathfrak{g})(k+h^\vee)}{2h^\vee}. \quad \checkmark$$

At $k = 1$: $\kappa = 9/4$. At $k = -2$ (critical): $\kappa = 0$. At $k = 0$: $\kappa = 3/2 = \dim(\mathfrak{sl}_2)/2$ (pure Sugawara shift; the classical r-matrix vanishes but the algebra still has three generators).

4.5. **Explicit trace: $\mathfrak{sl}_3$ at level k.** For $\mathfrak{g} = \mathfrak{sl}_3$: dim $= 8$, $h^\vee = 3$. The Lie algebra has the Chevalley basis $\{e_1, e_2, e_3, f_1, f_2, f_3, h_1, h_2\}$ (three positive roots, three negative roots, two Cartan generators), giving eight generators J^a for $a = 1, \ldots, 8$.

The r-matrix is $r(z) = k\,\Omega/z$ with $\Omega = \sum_{a=1}^8 t^a \otimes t^a$.

Double-pole channel. Each of the 8 generators has weight 1, moment factor $z^0 = 1$, and per-channel residue $\kappa_a^{\mathrm{dp}} = k$. Summing and normalizing:

$$\kappa_{\mathrm{dp}} = \frac{8k}{2h^\vee} = \frac{8k}{6} = \frac{4k}{3}.$$

Simple-pole channel. The adjoint Casimir eigenvalue is $C_2^{\mathrm{ad}} = 2h^\vee = 6$. The Sugawara trace:

$$\kappa_{\mathrm{sp}} = \frac{\dim(\mathfrak{g})}{2} = \frac{8}{2} = 4.$$

Total.

$$\kappa(V_k(\mathfrak{sl}_3)) = \frac{4k}{3} + 4 = \frac{4k+12}{3} = \frac{4(k+3)}{3} = \frac{\dim(\mathfrak{g})(k+h^\vee)}{2h^\vee}. \quad \checkmark$$

At $k = 1$: $\kappa = 16/3$. At $k = -3$ (critical): $\kappa = 0$.

4.6. **Computation: bc and $\beta\gamma$ systems.** The bc and $\beta\gamma$ systems have a distinct-weight structure ($h_b = \lambda \neq 1 - \lambda = h_c$ for $\lambda \neq 1/2$) and their only singular OPE is a cross-OPE.

The bc system. Generators b (weight λ) and c (weight $1-\lambda$). The cross-contraction has pole order 0 after d log-absorption:

$$\mathrm{Res}_{z=0}[r^{\mathrm{coll}}_{bc}(z)] = 0.$$

The collision residue has *no residue*. The kappa extraction from the r-matrix alone gives zero.

How does κ arise? For the bc system, the stress tensor is

$$T_{bc} = (\partial b)\, c \cdot \lambda + b\,(\partial c) \cdot (1 - \lambda),$$

a composite field of weight 2. The curvature $d_{\text{fib}}^2 = \kappa \cdot \omega_g$ arises not from the r-matrix directly (which is trivial for a simple-pole OPE) but from the Sugawara-type construction: the composite stress tensor generates a Virasoro subalgebra with central charge $c_{bc}(\lambda) = 1 - 3(2\lambda - 1)^2$.

The modular characteristic of the full bc vertex algebra is

$$\kappa(bc_\lambda) = \frac{c_{bc}(\lambda)}{2} = \frac{1 - 3(2\lambda - 1)^2}{2}.$$

This arises from the Virasoro subalgebra generated by T_{bc}: the bar complex of the bc system, as a factorization coalgebra, has fiberwise curvature controlled by the stress tensor, and $\kappa = c/2$ when the curvature is entirely generated by the Virasoro sector.

Boundary checks: $\lambda = 1/2 \Rightarrow c_{bc} = 1$, $\kappa = 1/2$ (single Dirac fermion). $\lambda = 2 \Rightarrow c_{bc} = -26$, $\kappa = -13$ (string reparametrization ghost).

The $\beta\gamma$ system. By complementarity $c_{\beta\gamma} + c_{bc} = 0$:

$$\kappa(\beta\gamma_\lambda) = \frac{c_{\beta\gamma}(\lambda)}{2} = \frac{2(6\lambda^2 - 6\lambda + 1)}{2} = 6\lambda^2 - 6\lambda + 1.$$

Checks: $\lambda = 1/2 \Rightarrow \kappa = -1/2$; $\lambda = 2 \Rightarrow \kappa = 13$; $\kappa(bc_\lambda) + \kappa(\beta\gamma_\lambda) = (1 - 3(2\lambda - 1)^2)/2 + (6\lambda^2 - 6\lambda + 1) = 0$ for all λ (Koszul complementarity, conductor $K = 0$).

Remark 4.3. The bc and $\beta\gamma$ systems illustrate a subtlety: when the only singular OPE is a *cross*-OPE (not a self-OPE), the collision r-matrix has no residue, and κ is not extracted by a naive trace-residue formula. Instead, κ arises from the Sugawara stress tensor of the composite system: the bar complex detects the conformal anomaly through the Virasoro sector, regardless of whether the generators have non-trivial self-OPEs. For free-field systems whose OPE is purely a cross-OPE, $\kappa = c/2$ exactly, where c is the central charge of the stress tensor. This is consistent with the general formula (4.2): the "Sugawara shift" subsumes the entire contribution when the diagonal r-matrix vanishes.

4.7. Degree-2: diagonal extraction rule.

Proposition 4.4. *For diagonal standard families, the scalar degree-2 contribution is extracted by the weight-dependent moment residue. For a single generator φ_i of weight h_i, the per-channel contribution is*

$$\kappa_i = \operatorname{Res}_{z=0}\left[z^{2h_i - 2} \cdot r_{ii}^{\text{vac}}(z)\right].$$

For abelian and Virasoro-type families, $\kappa(\mathcal{A}) = \sum_i \kappa_i$. For non-abelian affine Kac–Moody in the trace-form convention, $\kappa(\mathcal{A}) = \sum_i \kappa_i + \dim(\mathfrak{g})/2$.

4.8. Computation: Heisenberg. $r(z) = k/z$, $h = 1$.

$$\kappa_J = \operatorname{Res}_{z=0}\left[z^{2\cdot 1 - 2} \cdot k/z\right] = \operatorname{Res}_{z=0}[k/z] = k.$$

$$\boxed{\kappa(H_k) = k.}$$

Boundary checks: $k = 0 \Rightarrow \kappa = 0$ (uncurved); $k = 1 \Rightarrow \kappa = 1$.

4.9. Computation: Virasoro. $r^{\text{vac}}(z) = (c/2)/z^3$, $h = 2$.

$$\kappa_T = \operatorname{Res}_{z=0}\left[z^{2\cdot 2 - 2} \cdot (c/2)/z^3\right] = \operatorname{Res}_{z=0}\left[z^2 \cdot (c/2)/z^3\right] = \operatorname{Res}_{z=0}\left[(c/2)/z\right] = c/2.$$

$$\boxed{\kappa(\mathrm{Vir}_c) = c/2.}$$

Boundary checks: $c = 0 \Rightarrow \kappa = 0$; $c = 13 \Rightarrow \kappa = 13/2$ (self-dual point); $c = 26 \Rightarrow \kappa = 13$.

4.10. **Computation: $\mathcal{W}_N$.** Generators $W^{(s)}$ at weights $s = 2, \ldots, N$ with per-channel vacuum r-matrix $r^{\mathrm{vac}}_{ss}(z) = (c/s)/z^{2s-1} + \cdots$.

For each generator:

$$\kappa_s = \mathrm{Res}_{z=0}\left[z^{2s-2} \cdot (c/s)/z^{2s-1}\right] = \mathrm{Res}_{z=0}\left[(c/s)/z\right] = c/s.$$

Sum over generators:

$$\boxed{\kappa(\mathcal{W}_N) = \sum_{s=2}^{N} \frac{c}{s} = c \cdot (H_N - 1), \quad H_N = \sum_{j=1}^{N} \frac{1}{j}.}$$

Verification at $N = 2$ ($\mathcal{W}_2 = \mathrm{Vir}$): $\kappa = c \cdot (3/2 - 1) = c/2$.
Verification at $N = 3$: $\kappa = c \cdot (1/2 + 1/3) = 5c/6$.
Verification at $N = 4$: $\kappa = c \cdot (1/2 + 1/3 + 1/4) = 13c/12$.

Warning 4.5. $H_{N-1} \neq H_N - 1$. At $N = 2$: $H_1 = 1$ but $H_2 - 1 = 1/2$. The formula uses $H_N - 1$ (subtract the integer 1 from the N-th harmonic number), not H_{N-1} (the harmonic number at $N - 1$).

4.11. **Computation: affine Kac–Moody and the Sugawara shift.** For $V_k(\mathfrak{g})$ with $\mathfrak{g}$ simple, the r-matrix is $r(z) = k\,\Omega/z$ (trace-form convention). The averaging of the *classical* r-matrix alone does not produce the full κ. The discrepancy is the *Sugawara shift*: a one-loop self-contraction arising from the simple-pole OPE channel.

The proof proceeds in three steps.

Double-pole channel. The diagonal OPE $J^a(z)J^a(w) \sim k/(z-w)^2$ contributes to the fiberwise curvature d^2_{fib} on a genus-1 curve:

$$d^2_{\mathrm{fib}}\big|_{\mathrm{dp}}(J^a \boxtimes J^b) = (-2\pi i) \cdot k\,(J^a, J^b) \cdot \omega_1,$$

where $\omega_1 = c_1(\mathbb{E})$ is the Hodge class. Taking the trace over the normalized form $\sum_a (J^a, J_a) = \dim(\mathfrak{g})$:

$$\mathrm{tr}\big(d^2_{\mathrm{fib}}\big|_{\mathrm{dp}}\big) = k \cdot \dim(\mathfrak{g}) \cdot \omega_1.$$

Simple-pole channel (one-loop self-contraction). The cross-OPE $J^a(z)J^b(w) \sim f^{ab}{}_c J^c(w)/(z-w)$ feeds back into a second OPE contraction: J^c is contracted with itself through a second propagator loop, forming a one-loop Feynman diagram.

$$d^2_{\mathrm{fib}}\big|_{\mathrm{sp}}(J^a \boxtimes J^b) = (-2\pi i) \cdot \tfrac{1}{2} \sum_{c,d} f^{ac}{}_d f^{bc}{}_d \cdot \omega_1.$$

The factor $\frac{1}{2}$ is the symmetry factor ($|\mathrm{Aut}| = 2$ of the one-loop diagram). The contraction $\sum_{c,d} f^{ac}{}_d f^{bc}{}_d$ is the adjoint quadratic Casimir eigenvalue:

$$\sum_{c,d} f^{ac}{}_d f^{bc}{}_d = C_2^{\mathrm{ad}} \cdot (J^a, J^b) = 2h^\vee \cdot (J^a, J^b).$$

This is the Killing form identity: $K^{ab} = 2h^\vee\,\delta^{ab}$ in an orthonormal basis. Taking the trace:

$$\mathrm{tr}\big(d^2_{\mathrm{fib}}\big|_{\mathrm{sp}}\big) = \tfrac{1}{2} \cdot 2h^\vee \cdot \dim(\mathfrak{g}) \cdot \omega_1 = h^\vee \cdot \dim(\mathfrak{g}) \cdot \omega_1.$$

Combining the channels. On a single diagonal component, the two channels give per-component curvature $k + h^\vee$. Expressing the curvature as a multiple of the Casimir $\mathsf{C} = \sum_a J^a \otimes J_a$:

$$m_0 = \frac{k + h^\vee}{2h^\vee} \cdot \mathsf{C},$$

where $1/(2h^\vee)$ normalizes against C_2^{ad}. Then $\kappa = \mathrm{tr}(m_0)$:

$$\begin{aligned}\kappa &= \frac{k + h^\vee}{2h^\vee} \cdot \mathrm{tr}(\mathsf{C}) = \frac{k + h^\vee}{2h^\vee} \cdot \sum_a (J^a, J_a) \\ &= \frac{(k + h^\vee) \cdot \dim(\mathfrak{g})}{2h^\vee}.\end{aligned}$$

The decomposition:

$$\kappa = \underbrace{\frac{k \cdot \dim(\mathfrak{g})}{2h^\vee}}_{\kappa_{\mathrm{dp}} \text{ (tree-level, from } r(z))} + \underbrace{\frac{\dim(\mathfrak{g})}{2}}_{\kappa_{\mathrm{sp}} \text{ (one-loop, Sugawara shift)}}.$$

$$\boxed{\kappa\big(V_k(\mathfrak{g})\big) = \frac{\dim(\mathfrak{g})\,(k + h^\vee)}{2h^\vee}.}$$

Example 4.6 (Consistency checks).

$\mathfrak{g}$	dim	$h^\vee$	κ	$k = 1$	$k = -h^\vee$
$\mathfrak{sl}_2$	3	2	$3(k{+}2)/4$	$9/4$	0
$\mathfrak{sl}_3$	8	3	$4(k{+}3)/3$	$16/3$	0
E_6	78	12	$13(k{+}12)/4$	$169/4$	0
E_7	133	18	$133(k{+}18)/36$	$2527/36$	0
E_8	248	30	$62(k{+}30)/15$	$1922/15$	0
G_2	14	4	$7(k{+}4)/4$	$35/4$	0
F_4	52	9	$26(k{+}9)/9$	$260/9$	0

At $k = -h^\vee$ (critical level), $\kappa = 0$: the uncurved fixed point of Feigin–Frenkel. At $k = 0$, $\kappa = \dim(\mathfrak{g})/2$: the Sugawara shift alone.

4.12. **When the Sugawara shift vanishes.**

Proposition 4.7. *The Sugawara shift κ_{sp} is non-zero if and only if the algebra has multiple generators of the same conformal weight with non-trivial cross-OPE at the simple-pole level. In all other cases, $\kappa = \mathrm{av}(r(z))$ exactly.*

Three cases:

- (i) *Single-generator algebras* (Heisenberg, Virasoro): no cross-OPE, $\kappa_{\mathrm{sp}} = 0$.
- (ii) *Distinct-weight multi-generator algebras* ($\mathcal{W}_N$): the c-number contributions come only from diagonal self-OPEs; the cross-OPEs between generators of different weights have field-dependent leading coefficients that do not produce closed loops. $\kappa_{\mathrm{sp}} = 0$.
- (iii) *Same-weight multi-generator algebras* (Kac–Moody): all $\dim(\mathfrak{g})$ generators have weight 1, the structure constants $f^{ab}{}_c$ create the one-loop self-contraction, and $\kappa_{\mathrm{sp}} = \dim(\mathfrak{g})/2 \neq 0$.

4.13. **Summary of r-matrices and κ-values.**

Family	$r(z)$ (vacuum leading)	κ	κ_{sp}
H_k	k/z	k	0
Vir_c	$(c/2)/z^3$	$c/2$	0
$V_k(\mathfrak{g})$	$k\,\Omega/z$	$\dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$	$\dim(\mathfrak{g})/2$
$\mathcal{W}_N$	$(c/s)/z^{2s-1}$ per spin	$c(H_N-1)$	0
bc_λ	0 in collision residue	$(1-3(2\lambda-1)^2)/2$	all from Sug.
$\beta\gamma_\lambda$	0 in collision residue	$6\lambda^2-6\lambda+1$	all from Sug.

5. The chiral curvature

The averaging comparison extracts the symmetric degree-2 shadow from the ordered r-matrix. This is κ on abelian and scalar families after the appropriate moment-residue rule; for non-abelian affine Kac–Moody the averaged scalar is κ_{cl} and the full modular characteristic is $\kappa = \kappa_{\mathrm{cl}} + \dim(\mathfrak{g})/2$. The question it forces is: what role does κ play geometrically? The answer: κ is the proportionality constant of a curvature. The bar differential, which squares to zero on $\mathbb{P}^1$, fails to square to zero on higher-genus curves; the failure is proportional to κ. This section derives that formula from the propagator on a genus-g curve.

5.1. Propagators at higher genus.

On $\mathbb{P}^1$, the propagator $\eta_{12} = d\log(z_1 - z_2)$ is single-valued and the bar differential satisfies $d^2_{\mathrm{bar}} = 0$. On a genus-g Riemann surface Σ_g with $g \geq 1$, the function $z_1 - z_2$ is not globally defined. The single-valued (Arakelov-normalized) propagator is

$$\eta^{(g)}_{ij} = \Big[\partial_{z_i}\log E(z_i, z_j) + \pi \sum_{\alpha,\beta=1}^{g} \omega_\alpha(z_i)\,(\mathrm{Im}\ \Omega)^{-1}_{\alpha\beta}\ \mathrm{Im}\Big(\int_{z_0}^{z_j}\omega_\beta\Big)\Big](dz_i - dz_j),$$

where $E(z,w)$ is the prime form, $\omega_1, \ldots, \omega_g$ are the normalized abelian differentials, and Ω is the $g\times g$ period matrix.

At genus 1 ($\Sigma_g = E_\tau$ with period τ):

$$\eta^{(1)}_{12} = d\log\vartheta_1(z_1 - z_2 \mid \tau) + 2\pi i\,\frac{\mathrm{Im}(z_1 - z_2)}{\mathrm{Im}(\tau)}\,d\bar{z}_1,$$

where ϑ_1 is the odd Jacobi theta function. The non-holomorphic correction is forced by double-periodicity.

5.2. The curvature formula.

Theorem 5.1 (Fiber curvature and base obstruction)**.** *The fiberwise bar differential on $B(\mathcal{A})$ over Σ_g satisfies*

$$d^2_{\mathrm{fib}} = \kappa(\mathcal{A})\cdot\omega^{\mathrm{Ar}}_g, \tag{5.1}$$

where ω^{Ar}_g is the Arakelov $(1,1)$-form on the fiber curve. The corresponding class on the moduli base is obtained by the GRR/Deligne-pairing pushforward and is $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A})c_g(\mathbb{E})$.

At genus 1: $\omega_1 = \frac{i}{2\,\mathrm{Im}(\tau)}\,dz\wedge d\bar{z}$. At genus $g \geq 2$: ω_g involves the full period matrix $(\mathrm{Im}\ \Omega)^{-1}$, which is a $g\times g$ matrix (not a scalar).

The curvature factorizes:

$$d^2_{\mathrm{fib}} = \underbrace{\kappa(\mathcal{A})}_{\text{algebra}}\cdot\underbrace{\omega_g}_{\text{geometry}}.$$

The first factor is intrinsic to the chiral algebra (a quasi-isomorphism invariant, independent of the curve). The second factor is intrinsic to the complex structure of the curve (independent of the algebra). This factorization is a structural feature of the chiral theory with no direct classical Chern–Weil analogue.

5.3. **Cardy–Arakelov test for κ.** The curvature formula (5.1) is algebraic: $\kappa(\mathcal{A})$ is extracted from the r-matrix via the averaging map, and ω_g is a geometric form on Σ_g. The Cardy–Arakelov identity exhibits the same κ as the Arakelov-normalised coefficient of the coincident-limit torus two-point function of the stress tensor, thereby providing an independent geometric derivation of κ from $\langle T(z)T(w)\rangle_\tau$.

Conjecture 5.2 (Cardy–Arakelov test for the Virasoro lane). *Let $\mathcal{A}$ be a chirally Koszul algebra whose scalar invariant is the Virasoro lane $\kappa = c/2$, and let $\Sigma_1 = E_\tau$ be an elliptic curve with Arakelov $(1,1)$-form $\omega_1 = \frac{i}{2\,\mathrm{Im}(\tau)}\, dz \wedge d\bar{z}$ ($\because \int_{E_\tau} \omega_1 = 1$). Let $\langle T(z)T(w)\rangle_\tau^{\mathrm{reg}}$ denote the torus two-point function regularised by subtracting the $c/2 \cdot (z-w)^{-4}$ holomorphic short-distance singularity and the Sugawara self-energy shift ΔT_{Sug}. Then the modular Virasoro-lane scalar satisfies*

$$\frac{c}{2} \;=\; \lim_{w\to z} \frac{\langle T(z)T(w)\rangle_\tau^{\mathrm{reg}}}{\omega_1(z)} \;=\; \mathrm{Res}^{(2)}_{w=z}\big(\langle T(z)T(w)\rangle_\tau\big)\,/\,\omega_1(z), \tag{5.2}$$

where $\mathrm{Res}^{(2)}_{w=z}$ denotes the coefficient of $(z-w)^{-2}\, dz \wedge d\bar{z}$ in the genus-1 Arakelov propagator expansion. The identity is independent of the insertion point z and of the choice of Arakelov trivialisation, and is covariant under $\tau \mapsto (a\tau+b)/(c\tau+d)$.

Remark 5.3 (Proof obligation). The missing step is an Arakelov/Faltings or Bismut–Gillet–Soulé derivation identifying the regularised torus stress-tensor two-point coefficient with the Virasoro lane $c/2$. Stress-tensor two-point functions do not recover the programme's full level-trace invariant for Heisenberg or affine Kac–Moody without additional current/OPE data.

Remark 5.4 (Per-family verification). Equation (5.2) is a Virasoro-lane test. For Heisenberg and affine Kac–Moody, the full scalar invariant is extracted from current/OPE trace data and the Sugawara shift, not from the stress-tensor two-point function alone. For principal $\mathcal{W}_N$, the claimed equality with $c(H_N - 1)$ is a separate family computation.

Remark 5.5 (Chiral Chern–Weil content). Equation (5.2) is the proposed Virasoro-lane chiral Chern–Weil form of Cardy's identity. On the classical side, a connection's curvature pairs against an invariant polynomial to yield a Chern class; on the chiral side, the Maurer–Cartan element $\Theta_{\mathcal{A}}^{E_1}$ restricted to Σ_1 pairs against the Arakelov form ω_1 to yield the Virasoro scalar. The full programme invariant still requires the family-specific extraction rule.

5.4. **The chiral Chern–Weil map.** As $[\Sigma_g]$ varies over $\overline{\mathcal{M}}_g$, the bar complex forms a family of curved cochain complexes. The fiberwise curvature defines a characteristic class:

Theorem 5.6 (Chiral Chern–Weil). *For a uniform-weight modular Koszul chiral algebra $\mathcal{A}$, the obstruction class*

$$\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g \;\in\; H^*(\overline{\mathcal{M}}_g) \tag{5.3}$$

is a characteristic class of the family of bar complexes over $\overline{\mathcal{M}}_g$, where $\lambda_g = c_g(\mathbb{E})$ is the top Chern class of the Hodge bundle.

At genus 1, the formula (5.3) is unconditional for all algebras (uniform-weight or not).

The chiral Chern–Weil map:

$$\underbrace{\Theta_{\mathcal{A}}^{E_1}}_{\text{MC element}} \xrightarrow{\text{degree-2}} \underbrace{r(z) \in \mathfrak{g}_{\mathcal{A}}^{E_1}}_{\text{“connection”}} \xrightarrow{\text{av}} \underbrace{\kappa \in \mathbb{C}}_{\text{“Chern number”}} \xrightarrow{\cdot\,\lambda_g} \underbrace{\mathrm{obs}_g \in H^*(\overline{\mathcal{M}}_g)}_{\text{char. class}}.$$

Remark 5.7 (Comparison with classical Chern–Weil).

Classical	**Chiral**
Connection ∇	MC element $\Theta_{\mathcal{A}}^{E_1}$
Curvature $F_\nabla \in \Omega^2(M;\mathfrak{g})$	$r(z) \in \mathfrak{g}_{\mathcal{A}}^{E_1}$; $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$
Inv. polynomial $P \in \mathrm{Sym}^k(\mathfrak{g}^*)^G$	Averaging map $\mathrm{av}\colon \mathfrak{g}_{\mathcal{A}}^{E_1} \to \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$
$\mathrm{CW}(P) = P(F^k)$	$\mathrm{av}(r(z)) = \kappa_{\mathrm{cl}}$
Bianchi $d_\nabla F = 0$	MC: $d\Theta + \frac{1}{2}[\Theta,\Theta] = 0$
Char. class $[P(F^k)] \in H^{2k}(M)$	$\mathrm{obs}_g = \kappa \cdot \lambda_g \in H^*(\overline{\mathcal{M}}_g)$
Independence of connection	κ is a qi-invariant
$H^*(BG) = \mathrm{Sym}^*(\mathfrak{g}^*)^G$	Shadow tower $(\kappa, \mathfrak{C}, \mathfrak{Q}, \ldots)$

Three structural features have no classical analogue: (i) the base curve X itself varies over $\overline{\mathcal{M}}_g$; (ii) the curvature factorizes as algebra $\times$ geometry; (iii) the characteristic classes are graded by genus g, not by degree of an invariant polynomial.

6. The shadow tower

The modular characteristic κ is the degree-2 invariant of the chiral algebra, extracted from the classical r-matrix by the averaging map. But the ordered MC element $\Theta_{\mathcal{A}}^{E_1}$ has components at every degree: the KZ associator r_3 at degree 3, the quartic coherence r_4 at degree 4, and so on. Each carries E_1 operadic indices; each has a Σ_n-coinvariant shadow. The sequence of these shadows is the *shadow tower*, and its depth classifies chiral algebras into four types.

6.1. The shadow algebra.

Definition 6.1 (Shadow algebra)**.** Let $\mathcal{A}$ be a chiral algebra with modular cyclic deformation complex $\mathrm{Def}_{\mathrm{cyc}}^{\mathrm{mod}}(\mathcal{A})$. The *shadow algebra* is

$$\mathcal{A}^{\mathrm{sh}} := H_\bullet\big(\mathrm{Def}_{\mathrm{cyc}}^{\mathrm{mod}}(\mathcal{A})\big),$$

equipped with the descended Lie bracket from the convolution bracket on $\mathrm{Def}_{\mathrm{cyc}}^{\mathrm{mod}}(\mathcal{A})$.

Proposition 6.2. *The convolution bracket has degree* 0 *and maps degree* $r_1\times$ *degree* r_2 *to degree* $r_1 + r_2 - 2$ (*by gluing two external legs*)*. It descends to a well-defined graded Lie bracket on* $\mathcal{A}^{\mathrm{sh}}$*, making* $(\mathcal{A}^{\mathrm{sh}}, [-,-])$ *a bigraded Lie algebra.*

Proof. The bracket $[-,-]$ on $\mathrm{Def}_{\mathrm{cyc}}^{\mathrm{mod}}(\mathcal{A})$ is a chain map: $d[f,g] = [df,g] + (-1)^{|f|}[f,dg]$. Cocycles map to cocycles; the bracket of a cocycle with a coboundary is a coboundary. The descended bracket inherits graded antisymmetry and the Jacobi identity. □

The shadow algebra carries a bigrading by degree $r \geq 2$ and genus $g \geq 0$:

$$\mathcal{A}^{\mathrm{sh}} = \bigoplus_{r\geq 2} \bigoplus_{g\geq 0} \mathcal{A}^{\mathrm{sh}}_{r,g}.$$

6.2. **Shadow projections.** The individual shadow invariants are the bigraded components:

- $\kappa(\mathcal{A}) = \mathcal{A}^{\mathrm{sh}}_{2,0}$: the *modular characteristic* (degree 2, genus 0).
- $\Delta_{\mathcal{A}} = \mathcal{A}^{\mathrm{sh}}_{2,*}$: the *spectral discriminant* (all genera at degree 2).
- $\mathfrak{C}(\mathcal{A}) = \mathcal{A}^{\mathrm{sh}}_{3,0}$: the *cubic shadow* (degree 3, genus 0).
- $\mathfrak{Q}(\mathcal{A}) = \mathcal{A}^{\mathrm{sh}}_{4,0}$: the *quartic shadow* (degree 4, genus 0), containing S_4.
- $\mathrm{Sh}_r(\mathcal{A}) = \mathcal{A}^{\mathrm{sh}}_{r,0}$: the degree-$r$ shadow.

Each shadow is the image under av of the corresponding E_1 datum:

$$(\kappa,\ \mathfrak{C},\ \mathfrak{Q},\ \ldots) = \mathrm{av}\big(r(z),\ r_3,\ r_4,\ \ldots\big).$$

6.3. **What the higher-degree objects are.** *Degree* 2: the E_1 datum is the r-matrix $r(z)$, the classical Yang–Baxter datum encoding the collision of two fields. Its image under av is the scalar κ.

Degree 3: the E_1 datum is the KZ associator $r_3(z_1, z_2)$, a function on $\mathrm{Conf}_3(X)$ satisfying the pentagon identity. Its image under av is the cubic shadow $\mathfrak{C}$: the irreducible triple-collision invariant.

Degree 4: the E_1 datum is the quartic R-matrix coherence r_4, satisfying the quartic Stasheff equation. Its image under av contains the quartic contact invariant S_4.

General degree r: the obstruction class at degree $r+1$ is

$$o_{r+1}(\mathcal{A}) = \big(D_{\mathcal{A}}\Theta^{\le r}_{\mathcal{A}} + \tfrac{1}{2}[\Theta^{\le r}_{\mathcal{A}}, \Theta^{\le r}_{\mathcal{A}}]\big)_{r+1} \ \in\ H^2(\mathcal{A}^{\mathrm{sh}}_{r+1,0}).$$

This is a Lie bracket of shadows at degrees $\le r$: the shadow algebra bracket connects different degree components and drives the MC recursion.

6.4. **The discriminant and the G/L/C/M classification.**

Definition 6.3. The *discriminant* of a chiral algebra $\mathcal{A}$ is

$$\Delta = 8\kappa \cdot S_4.$$

Theorem 6.4 (Depth classification)**.** *For the standard verified families, the tuple* (S_3, S_4, *termination*) *classifies the four coarse depth types:*

Class	*Depth*	*Shadow criterion*	*Prototype*	Δ
G (Gaussian)	2	$S_3 = S_4 = 0$	*Heisenberg*	0
L (Lie)	3	$S_3 \neq 0,\ S_4 = 0$	*Affine KM*	0
C (Contact)	4	$S_3 = 0,\ S_4 \neq 0,\ S_{r\ge 5} = 0$	$\beta\gamma$	$\neq 0$ *generically*
M (Mixed)	∞	*non-terminating*	Vir, $\mathcal{W}_N$	$\neq 0$

Remark 6.5. The discriminant $\Delta = 8\kappa S_4$ is downstream data. It detects the quartic/contact channel when it is defined; it is not by itself an archetype classifier.

Remark 6.6 (Analogy with classifying spaces). The G/L/C/M classification is the chiral analogue of classifying principal bundles by the type of their structure group. The number of independent shadow invariants (characteristic classes) determines the complexity of the chiral classifying space. Class G (one invariant, κ) is analogous to $BU(1)$ (one class, c_1). Class M (infinitely many invariants) is analogous to $BO(\infty)$ (infinitely many Pontryagin classes).

A structural difference: classical Chern classes are polynomials in a single curvature F (the degree-2 datum); chiral shadow invariants are averages of distinct higher-degree operations r_n. Each shadow captures genuinely new collision structure not determined by lower degrees, unless $\Delta = 0$ and the tower terminates.

Remark 6.7 (Shadow depth and Koszulness). Shadow depth measures how many shadow invariants are non-zero. Koszulness (bar cohomology concentrated in degree 1) is a separate property. All standard families are Koszul; the shadow depths differ. SC-formality ($m_k^{SC} = 0$ for $k \geq 3$) holds if and only if the algebra is class G.

7. Multi-weight structure and cross-channel corrections

The scalar formula $\mathrm{obs}_1 = \kappa \cdot \lambda_1$ is unconditional at genus 1 for every family. On the uniform-weight lane, $\mathrm{obs}_g = \kappa \cdot \lambda_g$ extends to all genera via clutching morphisms on $\partial\overline{\mathcal{M}}_g$ (an independent GRR derivation provides a second path). The forced question: does it hold for multi-weight algebras, where generators have different conformal weights? The answer is no. At genus 1 the formula remains exact, while at genus ≥ 2 a cross-channel correction appears. The first witness is $\mathcal{W}_3$ at genus 2, where $\delta F_2 = (c+204)/(16c)$. This section derives these results from the graph-sum structure of the genus expansion.

7.1. Per-channel decomposition.

Let $\mathcal{A}$ be a modular Koszul chiral algebra with strong generators $\varphi_1, \ldots, \varphi_r$ of conformal weights $h_1, \ldots, h_r$. Each generator has a per-channel modular characteristic κ_i: the contribution from the (φ_i, φ_i) self-OPE channel. The total is $\kappa(\mathcal{A}) = \sum_{i=1}^r \kappa_i$.

The genus-g free energy is computed by a graph sum over stable graphs Γ of genus g:

$$F_g(\mathcal{A}) = \sum_{\Gamma} |\mathrm{Aut}(\Gamma)|^{-1} \sum_{\sigma\colon E(\Gamma)\to\{1,\ldots,r\}} A_\Gamma(\sigma, \mathcal{A}),$$

where σ assigns a channel to each edge e of Γ.

Definition 7.1. A channel assignment σ is *diagonal* if $\sigma(e) = i$ for all e (all edges carry the same channel). It is *mixed* if at least two edges carry different channels.

7.2. The decomposition theorem.

Theorem 7.2 (Conditional multi-weight genus expansion)**.** *Assume the diagonal genus-g Faber–Pandharipande comparison and the mixed-channel boundary graph expansion of Volume I. With the notation above:*

(i) Per-channel universality. *The diagonal contribution satisfies*

$$F_g^{\mathrm{diag}}(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}},$$

where $\lambda_g^{\mathrm{FP}} = \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}$ *is the Faber–Pandharipande number (*B_{2g} *the Bernoulli number), for all* $g \geq 1$ *in the diagonal sector.*

(ii) Cross-channel decomposition. *The full free energy decomposes as*

$$F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}} + \delta F_g^{\mathrm{cross}}(\mathcal{A}),$$

where $\delta F_g^{\mathrm{cross}}$ *is a graph sum over mixed-channel boundary graphs of* $\overline{\mathcal{M}}_{g,0}$*.*

(iii) Genus-1 universality. $\delta F_1^{\mathrm{cross}} = 0$ *for all* $\mathcal{A}$*.*

(iv) Uniform-weight universality. *If* $h_1 = \cdots = h_r$*, then* $\delta F_g^{\mathrm{cross}} = 0$ *at all* $g \geq 1$*.*

(v) R-matrix scope. *No higher-genus mixed-channel* R*-independence is asserted here. The* R*-independence statement is restricted to diagonal terms and to genus* 1*.*

(vi) Genus-2 explicit formula. *At genus* 2*:*

$$\delta F_2^{\mathrm{cross}}(\mathcal{A}) = \delta F_2^{\mathrm{sun}}(\mathcal{A}) + \delta F_2^{\theta}(\mathcal{A}) + \delta F_2^{\mathrm{lt}}(\mathcal{A}) + \delta F_2^{\mathrm{bb}}(\mathcal{A}),$$

summing over four contributing boundary strata among the seven stable graphs of $\overline{\mathcal{M}}_{2,0}$*.*

Proof. (i) On a diagonal assignment $\sigma(e) = i$ for all e, the amplitude factors as $A_\Gamma(\sigma_i, \mathcal{A}) = A_\Gamma(\mathcal{A}_i)$ for the rank-1 algebra $\mathcal{A}_i$ with curvature κ_i. Each rank-1 component has a single generator and hence uniform weight. Summing:

$$F_g^{\mathrm{diag}}(\mathcal{A}) = \sum_{i=1}^{r} F_g(\mathcal{A}_i) = \sum_{i=1}^{r} \kappa_i \cdot \lambda_g^{\mathrm{FP}} = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}.$$

(iii) At genus 1, the unique boundary stratum has a single edge. A single edge admits only r channel assignments, each pure. No mixed-channel assignments exist on a single-edge graph. □

7.3. **Why cross-channel corrections arise at genus g >= 2.** Three ingredients are needed for a non-zero $\delta F_g^{\mathrm{cross}}$:

(a) *Multiple edges on a single graph*, so that different edges can carry different channels. This requires $g \geq 2$, since all genus-1 contributing graphs have a single edge.
(b) *Distinct weights* on different channels ($h_i \neq h_j$), so that the propagators along edges carry different channel weights in the universal propagator.
(c) *Non-trivial OPE mixing*: the vertices carry mixed-channel OPE structure constants not controlled by κ alone.

At genus 1, ingredient (a) fails: the single-edge graph cannot carry a mixed assignment. At uniform weight, ingredient (b) fails: all propagators are identical regardless of channel. For free fields, ingredient (c) fails: $m_k = 0$ for $k \geq 3$ at genus 0, so mixed-channel vertices carry no structure constants beyond the two-point function.

7.4. **Explicit computation: $\mathcal{W}_3$ at genus 2.** $\mathcal{W}_3$ has two generators: T (weight 2) and W (weight 3). Per-channel modular characteristics: $\kappa_T = c/2$, $\kappa_W = c/3$. Total: $\kappa = c/2 + c/3 = 5c/6$.

The four contributing boundary strata at genus 2 give:

$$\delta F_2(\mathcal{W}_3) = \frac{3}{c} + \frac{9}{2c} + \frac{1}{16} + \frac{21}{4c} = \frac{c + 204}{16c}.$$

This is non-zero for $c \neq 0, -204$; for positive central charge it is non-zero.

- At $c = 1$: $\delta F_2 = 205/16$.
- At large c: $\delta F_2 \to 1/16$ (the leading correction is a constant, independent of c).
- At $c = -204$: $\delta F_2 = 0$ (an isolated zero, not physically relevant).

The four terms arise from distinct boundary mechanisms:

- $3/c$: sunset graph with mixed-weight vertex carrying the T–W OPE coefficient.
- $9/(2c)$: theta graph with three edges at mixed weight assignments $(2, 2, 3)$, $(2, 3, 3)$, and permutations.
- $1/16$: a topological contribution from the difference in Euler characteristics of the weight-2 and weight-3 determinant line bundles over $\overline{\mathcal{M}}_2$, independent of c.
- $21/(4c)$: bridge-loop graph connecting a genus-1 vertex (single-edge loop at weight h_i) to a genus-0 vertex through an edge at weight $h_j \neq h_i$.

Corollary 7.3. *The scalar formula $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}$ fails at $g = 2$ for $\mathcal{W}_3$. This resolves the multi-generator universality question in the negative: the scalar formula is not universal for interacting multi-weight algebras.*

7.5. **Free-field exactness.**

Proposition 7.4 (Free-field exactness). *Let $\mathcal{A}$ be a free-field chiral algebra ($m_k = 0$ for all $k \geq 3$ at genus 0). Then $\delta F_g^{\mathrm{cross}}(\mathcal{A}) = 0$ for all $g \geq 1$. The scalar formula $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ is exact at all genera.*

Three independent mechanisms force this vanishing:

(1) The genus-0 OPE is purely quadratic, so mixed-channel vertices carry no structure constants beyond the two-point function.
(2) The Givental R-matrix is trivial for free fields, eliminating R-corrections at genus-0 vertices.
(3) The propagator factorization identity for free fields collapses the cross-channel graph sum to the diagonal.

8. The chiral classifying space

The preceding sections constructed each ingredient of chiral Chern–Weil theory individually: the r-matrix (the connection datum), the averaging map (the invariant polynomial), the curvature formula, the shadow tower (the characteristic ring), and the multi-weight correction. The forced question: is there a single geometric object that unifies all of these, the way the classifying space BG unifies classical Chern–Weil theory?

8.1. $B(\mathcal{A})$ **as classifying object.** The ordered bar complex $B^{\mathrm{ord}}(\mathcal{A})$ on $\mathrm{Ran}(X)$, equipped with the deconcatenation coproduct, is a factorization coalgebra. In the deformation-theoretic picture, it is the chiral analogue of the simplicial bar construction $[n] \mapsto G^n$ whose geometric realization is the classifying space BG.

The universal bundle is the twisted tensor product

$$\mathcal{E}^{\mathrm{univ}}(\mathcal{A}) = \mathcal{A} \otimes^{\tau} B(\mathcal{A}),$$

with differential combining the internal differential, the bar differential, and the twisting morphism $\tau\colon B(\mathcal{A}) \to \mathcal{A}$. Its acyclicity, $\Omega(B(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ (bar-cobar quasi-isomorphism on the Koszul locus), is the chiral analogue of the contractibility of EG.

8.2. **The universal connection.** The MC element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$ is the *universal connection* on the chiral classifying space. The MC equation

$$d\Theta_{\mathcal{A}} + \tfrac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$$

is the flatness/integrability condition. At genus 0, the universal connection is flat ($d_{\mathrm{fib}}^2 = 0$); the universal bundle is trivializable over $\mathcal{M}_{0,n}$.

At genus $g \geq 1$, the curvature $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$ obstructs flatness. The universal bundle becomes curved, with characteristic class $\mathrm{obs}_g = \kappa \cdot \lambda_g$ on $\overline{\mathcal{M}}_g$.

8.3. **The total system over moduli.** The total system is a family

$$\mathcal{E}_{\mathrm{total}} \longrightarrow B^{\Sigma}(\mathcal{A})\big|_{\mathrm{Sym}^{\bullet}(C_g/\overline{\mathcal{M}}_g)} \longrightarrow \overline{\mathcal{M}}_g,$$

where $C_g \to \overline{\mathcal{M}}_g$ is the universal curve, and $B^{\Sigma}(\mathcal{A})$ is the Σ_n-coinvariant (symmetric) bar complex over the symmetric product $\mathrm{Sym}^n(X) = \mathrm{Conf}_n(X)/\Sigma_n$.

The ordered bar $B^{\mathrm{ord}}(\mathcal{A})$ on $\mathrm{Conf}_n(X)$ carries the full R-matrix $r(z)$ as its connection datum. The symmetric bar $B^{\Sigma}(\mathcal{A})$ on $\mathrm{Sym}^n(X)$ retains only $\kappa = \mathrm{av}(r(z))$ (plus the Sugawara shift for non-abelian families): the averaging map is the passage from ordered to unordered configurations.

As $[X]$ moves in $\overline{\mathcal{M}}_g$:

(i) The bar differential depends on X through the prime-form propagator $d\log E(z,w)$ (weight 1).
(ii) At genus 0: flat ($d_{\mathrm{fib}}^2 = 0$), universal bundle trivializable.
(iii) At genus $g \geq 1$: curvature $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$; characteristic class $\mathrm{obs}_g = \kappa \cdot \lambda_g$.
(iv) At multi-weight, genus $g \geq 2$: the scalar formula acquires the cross-channel correction $\delta F_g^{\mathrm{cross}}$.

8.4. **Crystallization.** Classical Chern–Weil theory has three ingredients: a connection ∇, an invariant polynomial P, and the resulting characteristic class $[P(F^k)]$. Chiral Chern–Weil theory has the same three: the E_1 MC element $\Theta_{\mathcal{A}}^{E_1}$ (whose degree-2 component is the r-matrix $r(z)$), the averaging map $\mathrm{av}\colon \mathfrak{g}_{\mathcal{A}}^{E_1} \to \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ (which kills the E_1 operadic indices on the descent locus), and the obstruction class $\mathrm{obs}_g = \kappa \cdot \lambda_g \in H^*(\overline{\mathcal{M}}_g)$. What is new is the variation over moduli: the curvature $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$ factorizes as algebra × geometry, the shadow tower $(\kappa, \mathfrak{C}, \mathfrak{Q}, \ldots)$ replaces the finite-rank invariant ring of a compact group with an degree-graded sequence whose depth classifies chiral algebras (G/L/C/M), and the scalar formula fails at genus ≥ 2 for multi-weight algebras with explicit correction $\delta F_2(\mathcal{W}_3) = (c+204)/(16c)$.

9. Koszul conductors and scalar complementarity

The chiral Chern–Weil output $\kappa(\mathcal{A})$ also organises Koszul-complementary pairs. Two distinct invariants live on a Koszul-complementary pair:

$$K_c(\mathcal{A}) \ := \ c(\mathcal{A}) + c(\mathcal{A}^!)$$

and the scalar κ-complementarity $K^\kappa(\mathcal{A}) := \kappa(\mathcal{A}) + \kappa(\mathcal{A}^!)$. There is no single ratio converting K_c to K^κ: the affine line has $K_c = 2\dim\mathfrak{g}$ but $K^\kappa = 0$, while Virasoro, principal $\mathcal{W}_N$, and Bershadsky–Polyakov use family-specific normalisations. Trinity values: $K_c = 0$ for class G and abelian Kac–Moody, $K_c = 2\dim(\mathfrak{g})$ for non-abelian $V_k(\mathfrak{g})$ while $K^\kappa = 0$ on the affine line; $K_c = 26$ for Virasoro (so $K^\kappa = 13$ at self-dual $c = 13$, with $\varrho_{\mathrm{Vir}} = 1/2$); $K_c = 196$ for the Bershadsky–Polyakov algebra BP (the polynomial identity $\mathrm{K}_{\mathrm{BP}}(k) \equiv 196$ as a rational function in the level, in Arakawa's convention, with the canonical fixed point at $k = -3 = -h^\vee(\mathfrak{sl}_3)$, where $K_{\mathrm{BP}}^\kappa = 98/3$ and $\varrho_{\mathrm{BP}} = 1/6$); $K = 100$ for principal $\mathcal{W}_3$ (so $K^\kappa = 250/3$ with $\varrho_{\mathcal{W}_3} = 5/6 = H_3 - 1$). The tabulated central charges of the bosonic $\beta\gamma$ and fermionic bc ghost systems provide the universal base: $c^{\beta\gamma}(\lambda) = 2(6\lambda^2 - 6\lambda + 1)$ and $c^{bc}(\lambda) = 1 - 3(2\lambda - 1)^2$, whose sum vanishes identically.

Proposition 9.1 (Verified conductor lanes). *For the verified standard families, central-charge conductors and scalar modular conductors are family-specific quantities. There is no single BRST ghost object canonically attached to every family in the table.*

Proof. For affine $V_k(\mathfrak{g})$, the central-charge conductor and the scalar modular conductor are different quantities:

$$K_c(V_k(\mathfrak{g})) := c(V_k(\mathfrak{g})) + c(V_{-k-2h^\vee}(\mathfrak{g})) = \frac{k\dim\mathfrak{g}}{k+h^\vee} + \frac{(-k-2h^\vee)\dim\mathfrak{g}}{-k-h^\vee} = 2\dim\mathfrak{g},$$

whereas

$$K^\kappa(V_k(\mathfrak{g})) := \kappa(V_k(\mathfrak{g})) + \kappa(V_{-k-2h^\vee}(\mathfrak{g})) = \frac{\dim\mathfrak{g}}{2h^\vee}\big((k+h^\vee) + (-k-h^\vee)\big) = 0.$$

The ghost calculation records the central-charge lane; the vanishing scalar lane is the affine complementarity used by the shadow tower. For Vir_c: the (b,c) ghost system has $\lambda = 2$ giving $c^{bc}(2) = -26$; complementarity $\kappa(\mathrm{Vir}_c) + \kappa(\mathrm{Vir}_{26-c}) = c/2 + (26-c)/2 = 13 = -c_{\mathrm{ghost}}/2$ after the universal factor. For $\mathcal{W}_N$: the reduction from $\mathfrak{sl}_N$ contributes a tower of ghosts at weights $\{2, 3, \ldots, N\}$; summation reproduces the tabulated K-value. For BP, the polynomial identity $K_{\mathrm{BP}}(k) \equiv 196$ in Arakawa convention matches the central-charge conductor $K_c^{\mathrm{BP}} = 196$ at every k; the scalar conductor is $K^{\kappa}_{\mathrm{BP}} = 196/6 = 98/3$, and the fixed point $k = -h^\vee(\mathfrak{sl}_3) = -3$ is a pole with principal-value half $49/3$. $\square$

Proposition 9.2 (Scalar complementarity and moment residues at genus zero)**.** *On the verified diagonal standard families, the genus-zero extraction of the scalar invariant uses the moment-residue pairing*

$$\kappa(\mathcal{A}) \;=\; \sum_i \mathrm{Res}_{z=0}\bigl(z^{2h_i-2} r^{\mathrm{vac}}_{ii}(z)\, dz\bigr) \;+\; \kappa_{\mathrm{sp}},$$

with the affine Sugawara shift κ_{sp} *added separately. For a Koszul pair, the scalar conductor is then* $K^{\kappa}(\mathcal{A}) = \kappa(\mathcal{A}) + \kappa(\mathcal{A}^!)$.

Remark 9.3 (Epistemic tags and independent-verification conventions). Every r-matrix in this section carries an explicit level prefix: KM trace-form $r^{\mathrm{KM}}(z) = k\,\Omega/z$ vanishes at $k = 0$; Vir $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$ has the canonical cubic pole. Every κ value is taken from the landscape census. The scalar conductor values $\{0, 13, 98/3, 250/3\}$ are verified against two disjoint sources under the independent verification protocol: direct computation of $\kappa + \kappa^!$ and ghost tabulation after conversion from K_c to K^{κ} where a ghost description is available.

Remark 9.4 (Cross-volume bridge). The central and scalar conductor lanes are separate Vol I manifestations of the trace formalism. In Vol III, the scalar normalization reappears on Borcherds–Kac–Moody algebras as $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ with $N \in \{1, 2, 3, 4, 6\}$ the level of the Fricke-type paramodular form. A numerical equality with a Vol I scalar conductor is not asserted without specifying the Borcherds input Φ_N, the normalization, and the Vol I family being compared.

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